



Indian Electric Vehicle Market Opportunities and Challenges



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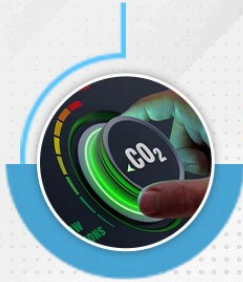
India is at the forefront of the global electric vehicle (EV) revolution, with **the mobility sector in emerging markets of South and Southeast Asia poised to absorb up to \$1.3 trillion in green capital by 2030**. This rapid growth presents both significant opportunities and challenges for the country.

Advantages of EVs in India

- 1. Lower Operating Costs:** Electric scooters and three-wheelers in India are currently around **\$40–\$112 cheaper** per year to own than petrol alternatives. This cost advantage is a major driver of adoption.
- 2. Reduced Emissions:** Mobility emissions represent **10% of total greenhouse gas (GHG) emissions** across Asia. Transitioning to EVs can significantly reduce air pollution and combat climate change.
- 3. Government Incentives:** The Indian government provides various incentives and subsidies to promote EV adoption, including reduced GST rates, income tax benefits, and financial incentives for purchasing EVs.
- 4. Technological Advancements:** The EV market is rapidly evolving, with continuous improvements in battery technology, charging infrastructure, and range capabilities. India is at the forefront of this innovation.
- 5. Domestic Manufacturing:** Indigenous companies like **Mahindra Last Mile Mobility** and **Battery Smart** are leading the push for electrification in India, creating jobs and boosting the economy.

THE BENEFITS THAT ELECTRIC VEHICLES IN INDIA OFFERS

LIMITING THE
CO₂ EMISSIONS



EASY
CHARGING



SIGNIFICANT COST SAVINGS &
REDUCED MAINTENANCE



MORE ECONOMICAL
THAN OTHER COMMERCIAL VEHICLES



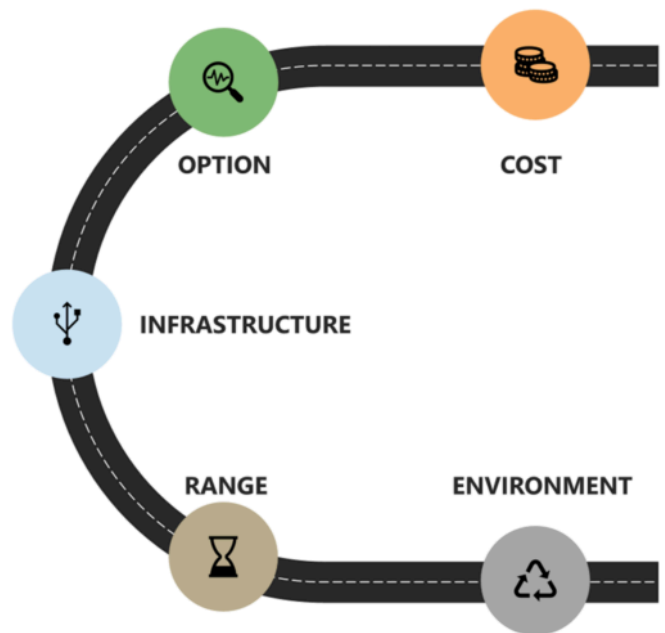
TROUBLE-FREE
DRIVING EXPERIENCE



Challenges Facing EVs in India

1. **High Initial Cost:** Despite government incentives, EVs generally have a higher upfront cost compared to conventional vehicles. This can be a barrier to adoption for many consumers.
2. **Limited Charging Infrastructure:** While India's charging infrastructure is growing, it is still relatively limited compared to petrol stations. Range anxiety and longer charging times can deter potential buyers.
3. **Battery Degradation and Replacement:** Over time, an EV's battery performance may degrade, resulting in reduced range and efficiency. Battery replacement can be expensive, although prices are expected to decrease as technology advances.
4. **Reliance on Imports:** India currently relies heavily on imports for critical EV components like lithium-ion batteries. Developing a domestic manufacturing ecosystem is crucial to reduce costs and ensure long-term sustainability.
5. **Potential for Higher Taxation:** As EV adoption increases, the government may need to raise taxes on electric vehicles to compensate for lost revenue from fuel taxes. This could impact the cost advantage of EVs.

THE CHALLENGES FOR ELECTRIC VEHICLES IN INDIA



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The Way Forward for India's EV Ecosystem

To fully capitalize on the opportunities presented by the EV revolution, India must address the challenges head-on. Key priorities include:

- Reducing the initial cost of EVs through targeted subsidies and incentives
- Expanding the charging infrastructure across the country
- Encouraging domestic manufacturing of EVs and components through supportive policies
- Investing in battery recycling and second-life applications to reduce reliance on imports
- Raising public awareness about the benefits of EVs and available incentives



By taking proactive measures and leveraging its strengths in technology and innovation, India can position itself as a global leader in the electric mobility space. The potential rewards are significant, both in terms of environmental impact and economic growth.

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